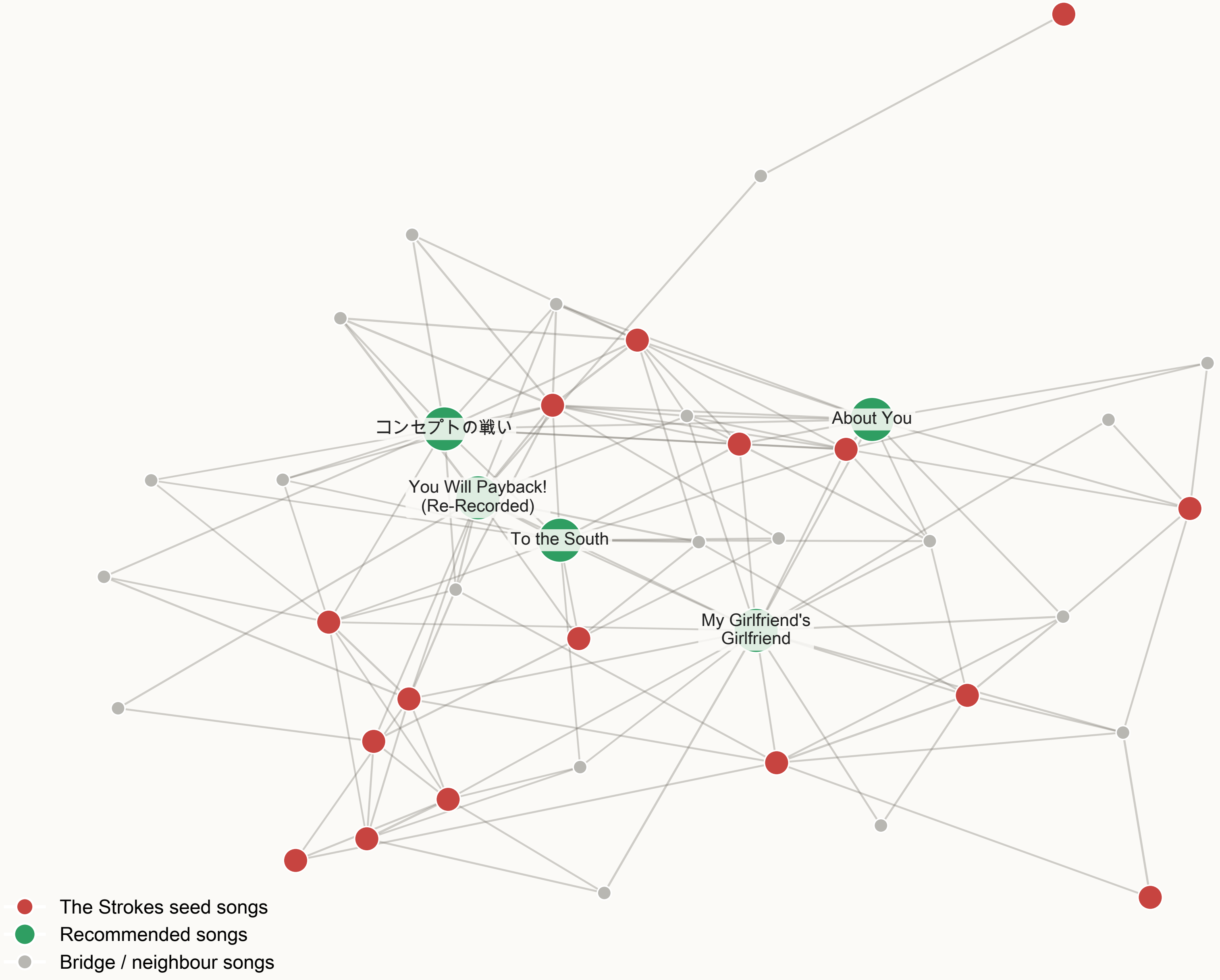
Spotify Graph Music Recommender — DS 4300

Personalized PageRank on a kNN audio-feature graph

Subgraph: The Strokes songs to top recommendations (<= 2 SIMILAR_TO hops)



Approach

The pipeline samples ~3,000 Spotify tracks stratified by track_genre, force-including every song by The Strokes and Regina Spektor. After z-scoring nine audio features (danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo), each song is connected to its K = 10 nearest neighbours by Euclidean distance, yielding a sparse SIMILAR_TO graph. Songs, artists, and genres are modelled as distinct nodes. Recommendation generation runs personalized PageRank (Neo4j GDS) over SIMILAR_TO, seeded with the user's liked songs and weighted by score = 1 / (1 + distance). The top-ranked tracks are returned after excluding artists the user already likes. The approach generalises to any user; Prof. Rachlin is just one seed set.

Graph Size

Song nodes: 3,045
Artist nodes: 3,252
Genre nodes: 114
Edges (SIMILAR_TO): 42,384 directed
Edges (unique undirected): 21,192
Density: (2 x 21,192) / (3,045 x 3,044) ~= 0.00457

Five Recommendations

#	Artist	Album	Track
1	Type O Negative	October Rust (Special Edition)	My Girlfriend's Girlfriend
2	Yuki Hayashi	Haikyu!! Karasuno vs Shiratorizawa OST	コンセプトの戦い
3	Confess	Back to My Future	You Will Payback! (Re-Recorded)
4	Motorama	Calendar	To the South
5	The 1975	Being Funny In A Foreign Language	About You

Similarity Rule

Two songs are connected by a SIMILAR_TO edge iff one is among the other's 10 nearest neighbours in z-scored audio-feature space, using Euclidean distance. Edges are symmetric and weighted by score = 1 / (1 + distance).